



XENIUS DocVault™

System Capabilities & Future Feature Catalog

"A comprehensive catalog of architectural upgrades, domain integrations, multi-modal enhancements, and autonomous capabilities that can be integrated into the XENIUS DocVault™ ecosystem."



Core Architectural & Infrastructure Enhancements



Async Event-Driven Ingestion Queue

Replace synchronous in-process OCR with an asynchronous task queue architecture (Celery + Redis / AWS SQS + Lambda). This unlocks:

- Non-blocking uploads (immediate API acknowledgment while OCR executes in background).
- Independent horizontal scaling of OCR/embedding workers.
- Automatic retry logic with dead-letter queues (DLQ) for malformed files.
- Real-time processing status streaming to the frontend via WebSockets or Server-Sent Events (SSE).

Recommended Stack: Celery, Redis, WebSockets, AWS SQS / EventBridge / Lambda



Multi-Tenant SaaS Architecture with BYOK Encryption

Transform the system into a multi-tenant platform with schema-level tenant isolation, Bring Your Own Key (BYOK) envelope encryption, tenant-scoped RBAC, and granular usage metering.

Recommended Stack: PostgreSQL RLS, AWS Secrets Manager, AWS KMS, SAML / OAuth2 SSO



Data Residency & Sovereignty Governance Engine

Build a geo-routing compliance layer enforcing data sovereignty standards (GDPR, India's DPDP Act 2023, CERT-In requirements). Enforces region-specific bucket storage, local SageMaker LLM inference, and verifiable audit compliance reports.

Recommended Stack: Geo-DNS Routing, AWS Local Zones, Regional KMS Keys



Knowledge Graph & Intelligence Layer Additions



GraphRAG — Knowledge Graph & Entity Extraction

Move beyond passage retrieval to entity-relationship extraction. Documents are parsed into nodes (*Person, Organisation, Feeder ID, Substation, Contract, Standard*) and directed edges (*signed_by, regulates, impacts, supersedes*). Complex questions like *"Which feeder standards were updated following Q3 outage reports?"* become structured graph traversals rather than imprecise text matching.

Recommended Stack: spaCy NER, Neo4j, LanceDB Graph API, NetworkX

OKF Studio — Visual Knowledge Base Manager

Deliver a browser-based interactive management studio for the Open Knowledge Format (OKF) featuring:

- Form-based YAML frontmatter builder (eliminates raw syntax errors).
- Live citation preview showing exactly how the entry will be synthesized in AI answers.
- Taxonomy management (add/rename/merge categories and alias conflict detection).
- Git integration: every save executes a signed Git commit directly to the knowledge repo.

Recommended Stack: [React](#), [Monaco Editor](#), [GitPython](#), [PyYAML](#)

Federated OKF & Cross-Vault Knowledge Synchronization

Extend OKF from local directories to a federated knowledge registry across multi-tenant deployments. Master utility instances publish verified metric definitions while subsidiary instances subscribe and sync changes automatically.

Recommended Stack: [REST Registry API](#), [Cryptographic Signatures](#), [Sync Engines](#)

Multi-Modal, Ingestion & Interface Additions

Multi-Modal Vision-Language Understanding

Extend Docling capabilities using Vision-Language Models (VLMs) to process charts, infographics, and complex single-line electrical diagrams (SLDs). Embedded visual assets are translated into structured, queryable data tables rather than treated as unparsed image blocks.

Recommended Stack: [Docling Vision](#), [Qwen2-VL](#), [LLaVA](#), [AWS SageMaker](#)

Corporate Communication Connectors (Email & Slack)

Build native ingestion adapters for enterprise communication channels (Microsoft Outlook/Exchange EWS API, Gmail Workspace API, Slack Bolt SDK, Microsoft Teams). Communication threads and attachments are ingested as RBAC-governed, searchable documents.

Recommended Stack: [EWS API](#), [Google Workspace API](#), [Slack Bolt SDK](#)

Real-Time Conversational Voice Interface

Integrate a low-latency speech-to-text (Whisper) → RAG → text-to-speech (Kokoro / ElevenLabs) pipeline. Field engineers on-site can query maintenance manuals, equipment schematics, and safety runbooks completely hands-free.

Recommended Stack: [Whisper](#), [Kokoro TTS](#), [WebRTC](#), [Mobile Edge Execution](#)

Domain Integrations & Autonomous Workflow Features

Localized Industrial Energy Management (EMS) Integration

Deep domain integration for energy sector document intelligence: direct ingestion adapters for OpenEMS, Battery Energy Storage System (BESS) maintenance logs, SCADA incident reports, and automated AT&C loss formula parsers comparing operational metrics against regional regulatory targets.

Recommended Stack: [OpenEMS Protocols](#), [SCADA Log Parsers](#), [Modbus/DNP3 Telemetry Parsers](#)

Agentic Document Workflow & Governance Engine

Deploy autonomous multi-step document agents capable of triggering lifecycle workflows: invoice auto-routing based on approval thresholds, scheduled legal-hold enforcement, and E-signature execution.

Recommended Stack: [LangGraph](#), [PydanticAI](#), [FastApi Background Tasks](#), [DocuSign API](#)

Predictive Document Intelligence & Anomaly Detection

Train ML models on access audit logs to predict upcoming document requirements prior to scheduled calendar events, flag retention schedule expirations, and detect anomalous access patterns signifying data exfiltration risks.

Recommended Stack: [Isolation Forests](#), [PyTorch](#), [Temporal Graph Models](#)